

DOOR-OPENING ALARM

■ T.K. HAREENDRAN

Children are at a high risk of drowning in a swimming pool. Majority of accidents occur when children get into a pool unsupervised. Such situations can be avoided by attaching a suitable alarm device to the door of the swimming pool. Here we describe the circuit of a 9V battery-operated electronic alarm driver that can be attached to the door leading directly to the pool. When the door is opened, the alarm sounds.

The circuit is built around resistor R2, a standard bar magnet, reed switch S2, IC CD4093 (N1 through N4), transistor T1 and some discrete components (refer Fig. 1). NAND gates

N1 and N2 are used as an inverter. An oscillator is built around gate N3, resistor R3 and capacitor C2. Gate N4 with resistor R4 and transistor T1 works as a buffer-cum-electromagnetic relay or buzzer driver.

When the door is closed, reed switch S2 is in closed state. When the door is opened, the bar magnet moves away from reed switch S2. As a result, the input to gate N1 is high. The low output (pin 3) of gate N1 makes the output of gate N2 high to enable the oscillator.

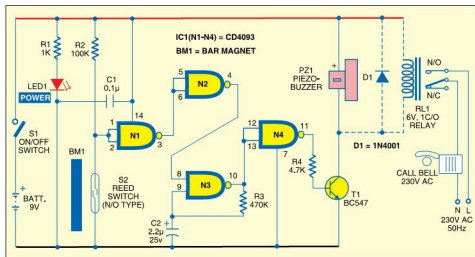


Fig. 1: Door-opening alarm circuit

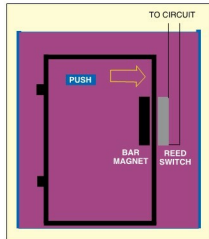


Fig. 2: Proposed door mounting arrangement

The output of the oscillator circuit is fed to piezobuzzer-driver transistor T1. When the door is opened, the piezobuzzer sounds an alarm alerting that someone is entering the swimming pool area.

Optionally, you can use electromagnetic relay RL1 (as shown with dotted line in Fig. 1) to operate a 230V AC mains operated call-bell or hooter. RL1 energises/de-energises at a frequency equal to that of the oscillator.

Assemble the circuit on a general-purpose PCB and enclose in a suitable cabinet. Connect the piezobuzzer at the back side of the cabinet. Install the unit on the swimming pool door as shown in Fig. 2. ●

METAL DETECTOR USING DIFFERENCE RESONATOR

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Described here is a simple circuit that can detect metallic conductors in its vicinity

up to a range of 25 to 30 millimetres. Concealed metallic objects such as metal foils enclosed in a plastic cover, e.g., toothpaste tubes, and small objects like refill tips made of magnetic

materials too can be detected using this circuit. However, very thin metallic foils may go undetected due to a large resistance.

The circuit is based on the principle

